

Review I

1. Section 1.1
 - a. Distance on a Number Line
 - i. Order Properties
 - b. Absolute Value
 - i. Relationship to Distance
 - ii. Properties
 - iii. Equations and Inequalities
 - c. Interval Notation
 - d. Distance in the Plane
 - e. Mid-Point Formula
 - f. Graph of an Equation
 - g. Equation of a Circle
 - h. Solving Trigonometric Equations
2. Section 1.2
 - a. Slope of a Line
 - i. non-Vertical Line
 - ii. Vertical Line
 - b. Forms of a Linear Equation
 - c. Parallel and Perpendicular Lines
 - i. Slope Criteria
 - d. Modeling: Best-Fit Line
3. Section 1.3
 - a. Definition of a Function
 - i. Domain, Range, Rule of Assignment
 - ii. One-to-One Functions
 - iii. Onto Functions
 - iv. Bounded Functions
 - v. Functional Notation
 - b. Piece-wise Defined Functions
 - c. Domain Convention
 - d. Composition of Functions
 - e. Classifications of Functions
 - i. Polynomial
 - ii. Rational
 - iii. Power
 - iv. Algebraic vs Transcendental
 - v. Trigonometric
 - vi. Exponential
 - vii. Logarithmic
4. Section 1.4
 - a. Graph of a Function
 - b. Vertical Line Test
 - c. Intercepts
 - i. x-intercepts (finding)
 - ii. y-intercepts (finding)
 - d. Symmetry
 - i. y-axis symmetry (finding)
 - ii. origin symmetry (finding)
 - e. Even and Odd Functions
 - f. Transformations of Functions
 - i. Translations
 - (1) Independent Variable
 - (2) Dependent Variable
 - ii. Scalings
 - (1) Independent Variable
 - (2) Dependent Variable
5. Section 1.5
 - a. Inverse Functions
 - i. Defined for One-to-One Functions
 - ii. Switches the Role of Domain and Range
 - iii. Inversion Formulas
 - (1) $f(f^{-1}(x)) = x$
 - (2) $f^{-1}(f(x)) = x$
 - iv. Horizontal Line Test
 - v. Solving an Equation for the Inverse
 - vi. Monotonic Functions
 - (1) Increasing / Decreasing
 - (2) Guarantees Existence of an Inverse
 - vii. Graphing the Inverse Function
 - (1) Reflection about the Line $y=x$
 - viii. Inverse Trigonometric Functions
 - (1) Restricting the Domain
 - ix. Trigonometric Identities Involving Inverse Functions
 - (1) Reference Triangles
 - (2) Reciprocal Formulas

6. Section 1.6
 - a. Properties of the Graph of $y = b^x$
 - i. Case $b > 1$
 - ii. Case $0 < b < 1$
 - iii. Monotonic
 - b. Properties of Exponents
 - c. Logarithm == Inverse of Exponential
 - d. Properties of the Graph of $y = \log_b x$
 - i. Case $b > 0$
 - ii. Case $0 < b < 1$
 - iii. Properties of Logarithms
 - e. Solving Exponential Equations
 - f. Solving Logarithmic Equations
 - g. Common and Natural Logarithms
 - h. Modeling: Exponential Growth and Decay

7. Section 2.1
 - a. Secant Lines
 - i. Difference Quotient
 - b. Tangent Line as Limit of Secant Lines

8. Section 2.2
 - a. Intuitive Notion of a Limit
 - b. Informal Definition of $\lim_{x \rightarrow c} f(x) = L$
 - c. One-side Limits, Two-Sided Limits
 - d. Estimating Limits Graphically
 - e. Estimating Limits using Tables
 - f. Infinite Limits $\lim_{x \rightarrow c} f(x) = \infty$
 - g. Formal Definition of $\lim_{x \rightarrow c} f(x) = L$

9. Representative Problems
 - a. 1.1 5, 13, 21, 29, 39
 - b. 1.2 3, 9, 21, 31, 35, 46, 50
 - c. 1.3 3, 8, 11, 16, 20, 23, 33, 47
 - d. 1.4 3, 9, 15, 28, 38
 - e. 1.5 4, 14, 20, 30, 38, 43
 - f. 1.6 3, 7, 15, 25, 39, 45-48, 60, 65, 68
 - g. 2.1 17, 29
 - h. 2.2 1-6, 8, 19, 27, 35, 53